Daily Drilling Report
SPD19B-03
4/8/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
14.0	16"	35.00	72.83	31.70	100.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	199,111.37	5,307.86	3,320,540.97		11.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
875.00	878.00		3.00	0.26	11.5

Operations at Report Time
Flushing 5 7/8" new DP.

Operations Summary
Tag 18 5/8" float collar @ 845m. D/O float collar and cement inside shoe track to 867.5m. Stuck pipe @ 867.5m after connection. Work on stuck and make it free. Cont' D/O shoe track and clear rathole to 875m. Drill 3m new formation to 878m. Perform FIT. POOH shoe track BHA#4 to surface. L/D 2 STD high rotating hours DP.

Operations next 24hrs
Make enough 8 1/4" DC and 5 7/8" DP. RIH 16" Directional BHA#5. Drilling 16" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Behnam Jahangiri	Drilling Engineer
Ali Sahebnouri	Geologist
M. H. Hemmati	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/9/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	3/9/2014	4/9/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	3/22/2014	5/22/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 31

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	03:00	3.00	16"	D	03	P	Drill out float collar @ 845m, cement inside shoe track to 867.5m w/ 900 - 1000 gpm/ 700 - 800 psi/ 5 - 30 kips/ 45 rpm/ 3 - 15 kft-lbf. ** Pump 40 bbl Hi-Vis pill after drill out float collar. ** Pump 40 bbl Hi-Vis pill while drilling @ 857m. ** Pump 40 bbl Hi-Vis pill when stand down @ 867.5m. ** Pull string inside float collar and check with & without rotation, Ok.
03:00	09:00	6.00	16"	D	24	N	(NPT for ABAN Company) After connection @ 867.5 , try to resume drilling, driller bring up both rig pumps SPM together to 45 in very short time. Suddenly pressure shoot up to 2100 psi. Then string start to move out of hole due to hydraulic up, lead to last DP STD bent and hit rig derrick structure. String got packed off and stuck. B/O bent DP with difficulty and disconnect TDS, R/B STD. Connect TDS. Try to move string down with 10 TRQ and 100 psi pressure, no success. Work on stuck by moving down (HL: 220 to 100 kips) while keep torque (10 - 15) inside string. no success. Increase TRQ to 30 and work on string. Release TRQ and keep 70 psi pressure, work on stuck by putting string in tension, at HL 375 kips, string start to move up and pressure decrease to zero, stuck got free. Increse RPM and GPM slowly, stablish normal rotation and circulation.
09:00	13:00	4.00	16"	D	07	P	Pump 30 bbl Hi-Vis pill. Circulate bottom up w/ 900 gpm/ 40 rpm/ 780 psi. Circulate and clear area f/ 867.5m to 840m. Spot 50 bbl Hi-vis pill before connection @ 867.5m.
13:00	14:00	1.00	16"	D	03	P	Cont' D/O cement inside shoe track f/ 867.5. Tag and D/O float shoe @ 870m. Drill and clear rathole to 875m. (800 gpm/ 650 psi/ 20 kips/ 40 rpm/ 4 - 12 kft-lbf) ** Pull string inside shoe and check with & without rotation, Ok. ** Pump 50 bbl Hi-Vis pill after drill out float shoe. ** Start displacing hole with 8.7 ppg simple Polymer mud while drilling.
14:00	15:30	1.50	16"	D	03	P	Drill 3m new formation f/ 875m to 878m w/ 800 gpm/ 650 psi/ 20 kips/ 40 rpm/ 3 - 8 kft-lbf, meanwhile displacing the hole: in & out: 8.7 ppg.
15:30	16:00	0.50	16"	D	11	P	Pull string inside shoe. L/U CMT unit to ckoke manifold, close Annular preventer on 5 7/8" DP. Perform FIT to 12.5 ppg EMW w/ 8.7 ppg simple Polymer mud by applying 550 psi surface pressure in steps, pressure drop to 400 psi in 5 min and get steady. Pump 8 bbls and get 6 bbls in return.

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Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
16:00	23:30	7.50	16"	D	04	P	POOH 16" rotary BHA f/ 878m to surface. R/B BHA. Check the bit (Green). ** L/D 57 jts 5 7/8" high rotating hours DP while POOH. ** L/D high rotating hour XOS (7 5/8" reg btm x 6 5/8" reg top) and replace with new one.
23:30	00:00	0.50	16"	D	20	P	P/U 2 STD high rotating hour 5 7/8" DP from derrick and L/D. >>> Offline job (16:00 - 17:00): Perform 26" CP x 18 5/8" annulus cement top job on wells B-05 and B-07. Well B-05 filled up after pumping 10 bbl 15.8 ppg neat cement and B-07 filled up after pumping 12 bbl 15.8 ppg neat cement. (Totally used 7 MT cement)
00:00	00:00						Apr 09 , 2014 00:00 - 06:00 P/U M/U and RIH 1 jt 9 1/2" NMDC + 9 1/2" Float sub + 8 jts 8 1/4" DC + 18 jts new 5 7/8" DP. Connect TDS and flush inside.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
878.00	Polymer Mud	8.70	50	19.0	21.000	6.0
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
6	8	0.0	95.0	5.0	23,030.000	
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	9.000		0/100		1245.0	1470.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
620.0	10.0	0.0			15.48739	0.418

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.20	4.	40	1.0	1,010.0	24.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Ahmed-06	Supply Boat	4/9/2014 04:20		Arrived from Aban 8, Delivering Cargo from SHJ.
Morshed	Supply Boat	4/8/2014 21:40		Delivered FW-200 MT, Delivering Cement-35 MT.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite BB	2.0	68.0	21.0
Bio-Cide	4.0	210.0	47.0
AMYLOSE B	40.0	400.0	121.0
Defoam SLC	1.0	220.0	79.0
Lime	1.0	338.0	140.0
H2S Scavenger	3.0	120.0	63.0
Bio-XCD	6.0	414.0	163.0
Caustic Soda	15.0	657.0	155.0
Pac-R	5.0	320.0	110.0
Pac-UL	15.0	160.0	54.0
Soda Ash	1.0	276.0	65.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	88.0
Drill Water	0.0	200.0	416.0
Bentonite Bulk	0.0	0.0	25.0
Bentonite BB	2.0	0.0	21.0
Cement G	7.0	0.0	32.0
Diesel Fuel	10.0	0.0	105.0
Potable Water	20.0	20.0	110.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
4	16	Varel	CR3GJMRS	255758	Mill Tooth	135	22/22/22/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.31	0.26	624.00	875.00	878.00	3.00	11.5	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
800	800	650.0	650.0	20	20	40	40
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
13.53	0.26	100.00	0.00				

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
875.00	878.00	0.26	5.0	800	40	650.0	20	

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
6 STD 5 7/8" HWDP	5 7/8	4.250	167.40	201.19
XOS	7	2.810	0.79	33.79
2 jts 8 1/4" DC	8 1/4	2.810	18.90	33.00

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4/8/2014

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Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
XOS	9 1/4	3.000	1.12	14.10
1 jt 9 1/2" DC	9 1/2	3.000	9.45	12.98
15 3/4" Stabilizer	15 3/4	3.500	2.17	3.53
Bit sub with float valve inside	9 1/8		0.95	1.36
16" Mill Tooth Bit	16		0.41	0.41

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.24	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.3

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29
Ilam		847.00	821.85

General Remarks

Comment

>>> Requirements: (150 MT Cement)

>>> Sent: NIL

>>> Received:

** Ahmed-06 from SHJ: (2 x Food Container)

** Morshed from Kish: (200 MT DW)